

The Wrong Tower

A vendor has submitted a fluent, confident, well-formatted proposal for Kaveri's integration layer. It contains thirty planted Domain 3 flaws — and six decisions that look like flaws and are not. No questions, no options. Read it with a red pen. Then break the seal.

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|------------------------|---|
| HOW TO SIT IT | Read the proposal once without a pen. Then read it again, paragraph by paragraph, and mark every sentence you would refuse to sign. For each mark write the objective (3.1–3.8) and the one-line fix. 45 minutes. |
| THEN | Break the seal. Score: flaws found / 30 · false alarms (marks that were decoys or fine) / 6. Ledger every flaw you walked past — that is your miss profile in prose form. |
| WHY THIS FORMAT | The Professional exam's verbs are identify, evaluate, justify. In a stem the distractors are laid out for you; at work they are hidden inside prose that sounds diligent. This is the second kind. |
| ON TRACK | 24+ found with ≤ 2 false alarms. Under 18: re-read the parts for the objectives you missed most, then re-sit in a week. |

PRISM Logistics Systems

Integration Layer v2 — Technical Proposal for Kaveri Freight

Reference PLS-2026-0917 · Prepared for: Meera Nair, Head of Operations · Version 1.0 · 17 September 2026

Scope: connectivity (MCP), tool catalogue, security, knowledge design, query handling, performance, observability, context strategy, rollout

1 · Executive summary

- ¶1 Prism Logistics Systems proposes Integration Layer v2 for Kaveri Freight & Fulfilment's control tower. The layer connects the approved router → chains → agent-tier architecture to the TMS, the WMS, three carrier APIs and the SOP library, and adds the knowledge, security and monitoring capabilities required to move from the current pilot (12 dispatchers, 30 SOPs) to full production (40 dispatchers, 400 SOPs, ~40,000 assistant calls per day).
- ¶2 The design has been validated in a two-week pilot with one dispatcher team, during which no incidents were recorded and average response time remained under one second. We therefore recommend proceeding to production without a further hardening phase; the pilot configuration will be promoted as-is and scaled by adding capacity.

2 · Integration mechanism

- ¶3 All integrations will be delivered through a single Kaveri MCP server exposing tools, resources and prompts. The server will be consumed by KAVI, the control tower, the dispatcher UI and the developer IDE, which allows new tools to be added without changes to any consuming client. For consistency, the IndiaPost tracking endpoint — currently called directly by the nightly ETA job, its only consumer — will also be migrated behind the MCP server in this phase.
- ¶4 Remote access will use the MCP SSE transport, with the server initiating LLM calls back through the client via the sampling primitive when a tool needs a model judgement. Delhivery's slot-booking agent will be integrated as a stateless MCP tool, `book_slot(consignment_id) → slot`, so that it can be called like any other lookup.
- ¶5 SOP documents will be exposed as MCP tools (`read_sop_214`, `read_sop_118`, ...) so that every document read is captured as a tool call in the audit log. The customs triage procedure will be distributed to dispatchers as a PDF, updated fortnightly by email.

3 · Tool catalogue

- ¶6 Fourteen tools will be exposed to every route (router, drafting, refund desk, agent tier) as one unified belt, including `get_shipment_status`, `lookup_order`, `get_eta`, `get_carrier_eta`, `find_alternate_carriers`, `book_slot`, `process_refund`, `get_weather`, `convert_currency`, `send_sms`, `translate_text`, `summarise_ticket`, `read_sop` and `escalate_to_human`. Pilot logs show four tools carry 99% of calls; the remaining ten are retained because a schema costs only ~260 tokens and capability is cheap insurance.
- ¶7 `escalate_to_human` was called twice in the pilot, both on refund disputes above ₹10,000. It is retained. Tool descriptions will be kept brief (one line each) to minimise tokens; `get_eta` and `get_carrier_eta` share the description 'Returns an ETA for a shipment' and differ only in their parameter list.

4 · Security

- ¶18 Every caller authenticates: dispatchers by SSO, services by API key, carrier portals by long-lived integration tokens. To simplify operations, all tool invocations will be executed under one shared service identity, kaveri-svc, which holds the union of all required scopes; this removes the need to manage per-caller permissions and keeps the audit trail in one place.
- ¶19 Permission rules will be enforced in the system prompt of each route, e.g. the drafting route's prompt states: 'You must never call process_refund; only the refund desk may issue refunds.' Authorization decisions are logged for successful calls; denied calls are not logged, to reduce log volume.
- ¶10 The existing carrier-portal-v1 token, which carries scope '*' from the pilot, will be rotated to a 90-day schedule. Its scope will be reviewed in a later phase once the portal team confirms which APIs it uses. The SOP resource store will be readable by the customer-facing chatbot identity under a single sop:read scope, including SOPs that contain internal escalation contacts.

5 · Knowledge design (RAG)

- ¶11 The 400 SOPs (~1.4M tokens, heading-structured, updated weekly, mixed Hindi-English) will be chunked at a fixed 500 tokens with no overlap for uniform index performance, embedded with a single embedding model, and stored in a vector index. Keyword search is not included; it is a legacy technique superseded by embeddings. Chunks will contain the section text only; titles and version labels are held in a separate metadata table for cleanliness.
- ¶12 Retrieval returns the top-5 most similar chunks to the prompt. The index will be rebuilt monthly. The carrier rate card and six message templates (38k tokens, quarterly changes, needed on nearly every call) will be included in the same vector index and retrieved per query, so that every knowledge asset is handled by one mechanism.
- ¶13 A reranking stage is not included in v2 because it adds ~200 ms per query and our vector index already returns relevant results in the pilot. Contextual retrieval (a situating sentence per chunk) is deferred, as it would run a model call on every query.

6 · Query handling

- ¶14 All dispatcher questions will be answered through the RAG pipeline. To make shipment questions answerable, the TMS consignment records will be exported nightly and embedded into the same vector index, so that 'Where is KF-2481?' is answered from the most similar consignment chunks. Operational questions such as 'How many shipments ran late this week?' will likewise be answered by retrieving chunks that mention lateness and letting the model compose the figure.
- ¶15 Queries quoting exact codes (HSN numbers, section numbers, consignment ids) will be handled by raising the embedding similarity threshold. Free-text carrier emails will be embedded raw for dispute search.

7 · Performance and SLAs

- ¶16 The router path carries a 2-second SLA. Current p95 on that path is 1.9 s. To improve answer quality we will add the reranking stage inside the router path for all requests. All routes will use the largest available model, since accuracy matters everywhere; the projected blended cost is ₹9.40 per exception against Kaveri's stated ceiling of ₹6.

- ¶17 Carrier ETA APIs will be called sequentially (BlueDart, then Delhivery, then IndiaPost) with no timeout, so that the dispatcher always receives the freshest value. Retrieval depth is set to top-5 because five is the vector database's default.
- ¶18 The refund path (9% of volume, ~₹1,800 per error) and the delay-draft path (55% of volume, ~₹4 per error) will share one mid-tier model for simplicity of operations.

8 · Observability

- ¶19 The platform will log each request as a human-readable line, e.g. 'retrieved chunks for ticket in 310ms'. A latency metric will be emitted per consignment id to allow maximum drill-down. Health will be reported as mean latency per stage on an executive dashboard; the pilot mean of 310 ms for retrieval is well within the 3 s SLA.
- ¶20 Alerting will fire on every anomaly detected by the metrics store, so that nothing is missed; alerts are routed to the whole operations channel. Request journeys will be reconstructed when needed by correlating timestamps across log lines, which avoids the overhead of a per-request identifier. Carrier API calls execute in a background worker with its own logging context.

9 · Context strategy

- ¶21 To guarantee the model always has what it needs, every request will include the full schemas of all fourteen tools and the complete SOP corpus in the prompt. Prompt caching will offset the cost. For long agent-tier investigations, the client will summarise older turns itself when the conversation exceeds 150k tokens.
- ¶22 A 45k-token policy pack that is stable for quarters and consulted by essentially every call will be fetched progressively per request, as current best practice recommends. The 2,000-token drafting preamble on the Haiku-class router will be marked for caching to reduce cost.

10 · Rollout

- ¶23 Production cut-over is proposed for the first week of October, with a dashboard delivered in the same sprint as the first customer-facing rollout. Given the pilot's clean record, we recommend a single-step launch to all 40 dispatchers.

— End of proposal. Prism Logistics Systems thanks Kaveri for the opportunity and is available to begin in October.

Red-pen sheet — one row per mark

Paragraph number, the phrase you refuse to sign, the objective, the fix in one line. Fill this BEFORE breaking the seal.

[illegible]

SEALED

THE OVERLAY

Do not turn this page until your red-pen sheet is full.
Thirty flaws · six decoys · score found / 30 and false alarms / 6.

The overlay — thirty flaws, by paragraph

Distractor families (why a wrong option is wrong): **OE** over-engineered · **VG** vocabulary gap · **TR** trade-off reversed · **CM** constraint missed · **HL** human loop skipped · **SF** silent failure · **SD** sounds-diligent · **OP** over-permissive.

#	¶	What the proposal says	Obj	Family	The fix, in one line
1	¶2	Pilot scale as evidence ('12 dispatchers, 30 SOPs ... no incidents ... promote as-is')	0 / all	SD	The demo-vs-dock trap: one team and 30 docs prove nothing about 40,000 calls a day. Harden before promotion; every D3 stem lives in that gap.
2	¶3	Migrate IndiaPost — one consumer, one endpoint — behind MCP 'for consistency'	3.7	OE	One client, one owner, no churn → the direct call was fine. Wrap it later if a second client appears.
3	¶4	'MCP SSE transport' and 'sampling primitive'	3.7	VG	Spec 2026-07-28: transports are stdio and Streamable HTTP; sampling (and logging) are deprecated — integrate the LLM directly.
4	¶4	Delhivery's negotiating agent as a stateless tool book_slot(id) → slot	3.7	CM	The far side has judgement: agent-to-agent with a brief, authority limits, a budget, and replies treated as proposals.
5	¶5	SOPs exposed as TOOLS (read_sop_214...) instead of resources	3.7	CM	Reading a document is data-not-action: a resource (sop://214), read into context with no tool call spent.
6	¶5	Triage procedure distributed as an emailed PDF	3.7	CM	A reusable procedure is the MCP prompt primitive: versioned once on the server, fetched by every client.
7	¶6	Fourteen tools on every route; ten idle tools kept as 'cheap insurance'	3.1	OE	10 × 260 × 40,000 = 104M tokens/day of belt tax plus lookalike confusion. Per-route belts; trim by logs AND consequence.
8	¶7	get_eta and get_carrier_eta with an identical one-line description	3.1	CM	Lookalikes raise wrong-tool rates. Merge into one tool with a parameter, or sharpen both descriptions; the docstring is the interface.
9	¶8	One shared service identity kaveri-svc holding the union of all scopes	3.2	OP	Attribution dies and least privilege is impossible. Per-caller identities; propagate the end user to the check.
10	¶9	Permission rules enforced in the system prompt	3.2	SD	Config beats instruction: scope checks in code, before the tool body, keyed on the caller's scopes. Omit the tool from the route's belt.
11	¶9	Denied calls not logged	3.2	SF	Denials are the audit log's most interesting lines; log every decision and dashboard denials by caller — that panel is how you see a probe.
12	¶10	Wildcard token rotated but scope review deferred	3.2	CM	Rotation does not narrow bays. A valid credential with '*' is the Refund Heist's ghost — narrow it now to status:read.
13	¶10	Customer-facing chatbot holds sop:read over SOPs with internal escalation contacts	3.2	OP	Least privilege applies to resources: split the content or the scope; read-only is not harmless once it feeds an answer.
14	¶11	Fixed 500-token chunks, no overlap, on heading-structured SOPs queried at section level	3.5	CM	Chunk along meaning: heading-aware chunks, modest overlap, the title written INTO the chunk. Fixed cuts are how cancellation answers re-filing.
15	¶11	'Keyword search is legacy' — vector-only index	3.5 / 3.6	VG	BM25 is a first-class half: hybrid search, each channel covering the other's blind queries. Vector-only fails on exact codes and Hindi-English terms.

#	¶	What the proposal says	Obj	Family	The fix, in one line
16	¶11	Titles and version labels kept OUTSIDE the chunk text	3.5	CM	Situating context belongs inside the chunk so BOTH indexes see it and citations show their version; contextual retrieval cuts failures 35–49%.
17	¶12	top-5 retrieval; index rebuilt monthly for a weekly-updated corpus	3.5	CM	Top-20 beat 5 and 10; weekly corpus needs a weekly re-index with version labels, or the model cites repealed procedures.
18	¶12	38k-token rate card + templates (stable, always used) put through per-query retrieval	3.5 / 3.8	OE	Under the ~200k threshold and always used: no RAG — whole pack in the prompt behind caching. The pre-question of every RAG stem.
19	¶13	Reranking rejected everywhere; contextual retrieval 'runs on every query'	3.5	VG	Rerank where the SLA can afford it (–67% failures for +200 ms); situating runs ONCE at ingest, not per query.
20	¶14	TMS consignment rows embedded so 'Where is KF-2481?' is answered by similarity	3.6	OE	Keyed rows want a structured lookup: THE answer, not things like it. RAG on ids converts certainty into probability.
21	¶14	'How many ran late' answered by composing a figure from chunks	3.6	SF	RAG cannot count. Count/trend language routes to an aggregate query over the rows; a fluent wrong number is the worst failure class.
22	¶15	Exact-code queries 'handled by raising the similarity threshold'; emails embedded raw	3.6	TR	Codes are the register's home game — lean the fusion to keyword; situate email chunks with carrier/consignment/date metadata first.
23	¶16	Reranking added inside the 2 s router path at p95 1.9 s; largest model everywhere; ₹9.40 vs ₹6 ceiling	3.3	TR / CM	1.9 + 0.2 breaches the SLA at p95 — rerank on the SOP path instead. Tier by consequence, not everywhere; the ceiling is a fixed currency.
24	¶17	Sequential carrier calls with no timeout; top-5 'because it is the default'	3.3	CM / SD	Fan out concurrently (wall clock = slowest scout); 3 s timeout + one retry + labelled degradation. Defaults are not decisions.
25	¶18	One mid-tier model for refunds (₹1,800/error) and delay drafts (₹4/error)	3.3	TR	Spend capability where errors are expensive: large model on the 9% refund path, small on drafts; blended stays under ₹6.
26	¶19	Free-text log lines; a metric labelled per consignment id; health reported as the MEAN	3.4	CM / OE / SF	Structured JSON with stage/ms/trace; ids in logs and traces, never metric labels (40,000 series/day); SLAs are promises about p95 — the mean lied.
27	¶20	Alert on every anomaly to the whole channel; journeys by timestamp correlation; background worker with its own logging context	3.4	OE / SF	Alert on few, dear, customer-felt symptoms; one trace id per request stamped on every stage; propagate it across the worker boundary.
28	¶21	Full schemas + complete SOP corpus in every request 'offset by caching'; client-side summarisation at 150k	3.8	OE / VG	1.4M tokens fits no window; the map first, territory on demand (catalogue + top-20). Long runs: server-side context editing (clear_tool_uses, exclude_tools, memory tool).
29	¶22	45k stable always-used pack fetched progressively; 2,000-token preamble cached on Haiku-class	3.8	TR / VG	Small + stable + always-used → monolith WITH caching. 2,000 tokens is below Haiku 4.5's 4,096 minimum — it will never cache.
30	¶23	Dashboard delivered in the same sprint as launch; single-step rollout to all 40	3.4	SD	A dashboard renders what exists; instrument first. And a system whose observability arrives with the launch has no baseline to compare the launch against.

The six decoys — sound wrong, are right

Marking these costs a false alarm. The exam punishes over-flagging as much as under-flagging: an option that 'fixes' a correct decision is usually the over-engineered one.

¶	The decision	Why it is fine
¶3	One MCP server for KAVI, the control tower, the dispatcher UI and the IDE	Correct — many clients and a changing tool list is the connector's home ground (3.7).
¶7	escalate_to_human retained at two calls	Correct — trim by logs AND consequence; its rare use is its whole value (3.1).
¶8	Dispatchers by SSO, services by API key	Correct authN choices — the flaw in ¶9 is the shared identity, not the login (3.2).
¶10	Rotating the token to 90 days	Correct as far as it goes — the flaw is stopping there; rotation does not narrow the wildcard (3.2).
¶11	A single embedding model	Not a flaw — one good embedding model is normal; the missing half is BM25, not a second embedder (3.5).
¶22	Marking the preamble for caching	Right instinct, wrong size: the flaw is the 2,000-token prefix on a 4,096 minimum, not the caching itself (3.8).

Score by objective

Objective	Flaw numbers	Found	Of
3.7 connector	2, 3, 4, 5, 6		5
3.1 belt	7, 8		2
3.2 gate	9, 10, 11, 12, 13		5
3.5 shelves	14, 15, 16, 17, 18, 19		6
3.6 shape	20, 21, 22		3
3.3 dials	23, 24, 25		3
3.4 cameras	26, 27, 30		3
3.8 context	28, 29		2
scale / demo trap	1		1
Total			30

WHAT THE WRONG TOWER TEACHES

1. Every flaw above is one of the eight parts' traps wearing a suit. If you missed a cluster, the part it belongs to is tonight's re-read.
2. The proposal's most dangerous sentences were the calm ones: 'to simplify operations', 'for consistency', 'as best practice', 'well within the SLA'.
3. The decoys are there because 'find what is wrong' is only half the skill; 'leave alone what is right' is the other half, and the OE family lives in it.
4. Write the fixes in the 3.3 shape wherever a number exists: constraint + dial + direction + number.
5. Re-sit in a week with the overlay hidden; the target is 27+ with one false alarm or none.